ELECTRICAL ENGINEERING DEPARTMENT
COLORADO SCHOOL OF MINES

EENG 385 - Electronic Devices and Circuits
Frequency Domain: Active Filters
How To: Use Simulator to Build a Bode Plot Using Point-by-Point method

How To: Simulation Bode Plot – Point by Point

Let’s construct the low pass filter (LPF) in Figure 1 using LTspice.

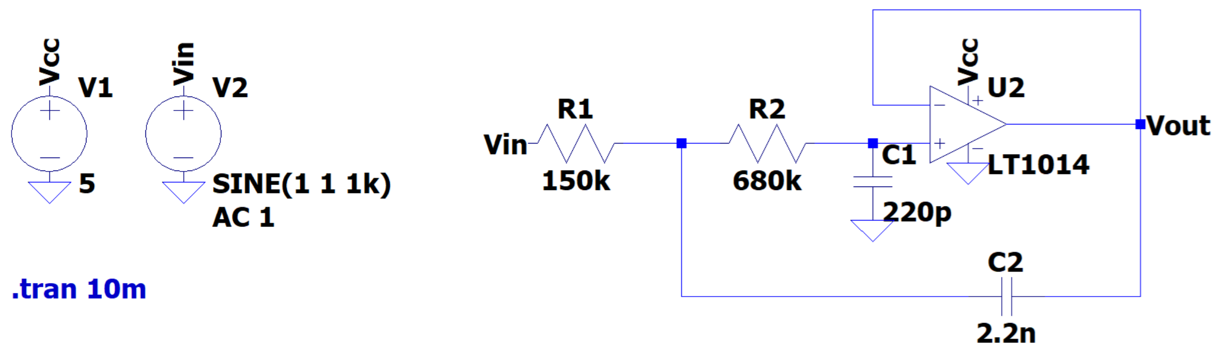

Figure 1: The low-pass filter on the Audio board is built around an op amp powered by 5V and gnd.

The 5V DC supply, V1, for the op amp is an important detail copied over from the real Audio board filter. Voltage source V2 provides a 1kHz sinusoid input to the LPF. When you place a voltage source, go to the Advanced options, select the SINE radio button and make the following :

- DC offset[V] 1
- Amplitude[V] 1
- Freq[Hz] 1k

The op amp in Figure 1 (and on your Audio board) is powered from 5V and 0V. For the LPF and the BPF this will cause problems if your input signal has no DC bias. “No DC bias” means that the input sine wave is centered at 0V - the audio input to your Audio board has no DC bias. In our simulation, the DC bias is provided by the 1V DC offset of V2. On your Audio board, the DC bias is provided by the level shifter.

With this out of the way, generate the waveforms in Figure 1. After probing Vin and Vout, your waveforms should be identical to Figure 2.

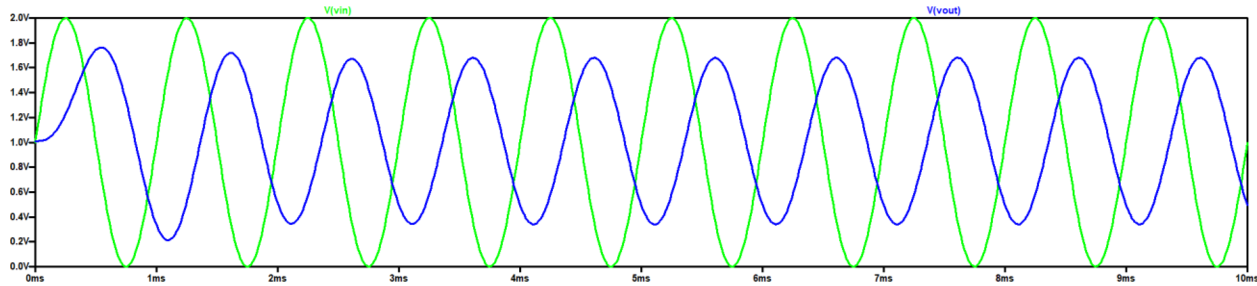

Figure 2: A 1,000Hz input waveform (in green) and the output(blue) from the low-pass filter.

From this graph, you should be able to see that the output waveform has a lower amplitude than the input. This change in amplitude, from input to output, is called attenuation. Mathematically, attenuation is the ratio of the output over the input. You will form this ration to normalize the output amplitude in terms of the input amplitude. Since we will be measuring very large ranges of attenuation, you will measure attenuation in units of decibels by using the following equation.

$$attenuation\ in\ decibels = 20 * \log({}^{V_{out}}/_{V_{in}})$$

Let's measure the attention, in decibels, of the waveforms in Figure 2.

- By observations, Vin is 2.00V peak-to-peak
- Using the cursors on the Vout, it is 1.36V peak-to-peak

Thus at 1,000 Hz, the low-pass filter has 20*log(1.36/2.00) = -3.3dB of attenuation.

In addition to the change in the amplitude, the output waveform is delayed with respect to the input waveform. This shift in time is called phase delay. Phase delay is measured in degrees that the output waveform occurs with respect to the input waveform. Let's measure the phase delay of the output in Figure 2. Start by choosing any peak of the input; I'll choose the peak at 2.25ms. Now look for the peak in the output that immediately follows the input. This occurs at 2.6ms. To find the phase angle, divide the time different between these two peaks by the period of the waveform; (2.25ms – 2.6ms)/1ms = -0.35. Finally, multiply this value by 360° to get the phase delay; -0.35*360° = -126°.

So, at 1,000Hz the low pass filter in Figure 1 has -3.3dB of attenuation and -126° of phase shift. A Bode plot is a pair of graphs describing the attenuation and phase shift of a circuit as a function of frequency. You can build a Bode plot by repeating the calculations in the previous paragraph multiple times. So, let's give it a try.

Start by downloading and opening the filterWorkSheet posted on the Canvas page. Go to the LPFsim tab and fill in value for V_{out}/V_{in} for 1000Hz row and enter the ratio of the output over the input; 0.75 in cell C9. The corresponding decibel entry will auto populate in the Attenuation column. Next enter the time delay between the peaks of the input and output waveforms in units of milliseconds in the Delta (ms) column; -0.35 in cell E9. The corresponding Phase entry will auto-populate in the Phase column.

Now change the frequency of the input waveform in LTspice to every frequency value in column B that is not already filled in. Some technical notes:

- When you get to higher frequencies, the amplitude of the output waveform will be so small that it becomes hard to measure. When this happens, zoom in with thin wide zoom rectangles.
- With the mouse pointing at an axis, you can translate along the axis using the cursor keys.
- If the output waveform will not stay centered at 1.0V, determine the amplitude over a single wavelength. Do you best.
- To measure the phase, I left one x-axis cursor at a peak of the input and then zoomed in on the vertical axis until I could move the other x-axis cursor to an output peak.

I'd like to take a moment and draw your attention to the frequencies you used as the input frequencies and listed in column B of the spreadsheet. Note that these frequencies follow a pattern of 1.0, 2.2, 4.7 before repeating at the next power of 10. As an electrical engineer you would say that we have 3 different frequencies per decade. These 3 frequencies are separated by a multiplicative factor of $\sqrt[3]{10} = 2.15 \ \approx 2.2$. When you plot numbers whose x values are separated by a multiplicative factor of 2.15 on a logarithmic horizontal axis the points appear equally separated along the axis. This is a result of the fact that $\log(2.15^n) = n * \log(2.15)$. So in short, using a geometric factor to increase frequencies ensures the points are space equally along a logarithmic axis. Neat huh?

You can usually copy graphs from Excel and paste them into Word. At worst you can use the windows snipping tool to make a copy of your Bode plot. Try to make the graph reasonably large so that the graders can read the units in the horizontal and vertical axes.

Now you are ready to complete the plots asked for in the assignment – good luck!